Notation
ECON2050

Below some notation that we use in the ECON2050 lecture slides. It is a short and incomplete list — I only added stuff that some of you or past students have asked me about.

(1) $\mathbb{R}$
Meaning: set of real numbers
Note: The textbook by Simon and Blume (SB) uses $\mathbf{R}$ or $\mathbf{R^1}$ to denote $\mathbb{R}$. See the beginning of Chapter 2.1 of SB for an additional explanation of real numbers.

(2) $\mathbb{N}$
Meaning: set of natural numbers, i.e. $\mathbb{N} = \{1, 2, 3, \ldots\}$

(3) $\in$
Meaning: "is an element of"
Example: $1 \in \{1, 2, 5\}$ means "1 is an element of the set $\{1, 2, 5\}$."

(4) $\forall$ (see also Tutorial 1)
Meaning: "for all"
Example: $\forall x > 1, x \in \mathbb{R} : x^2 > x$ means "for all real numbers x that are greater than one, $x^2 > x$ holds"

(5) $\exists$ (see also Tutorial 1)
Meaning: "exists"
Example: $\exists \varepsilon > 0 : \varepsilon < 1$ means "there exists a strictly positive number ε such that ε is strictly less than 1"

(6) $\subseteq$
Meaning: "is a subset of"
Example: $\{1, 2, 3\} \subseteq \mathbb{N}$ (true because any element of $\{1, 2, 3\}$ is also an element of $\mathbb{N}$)

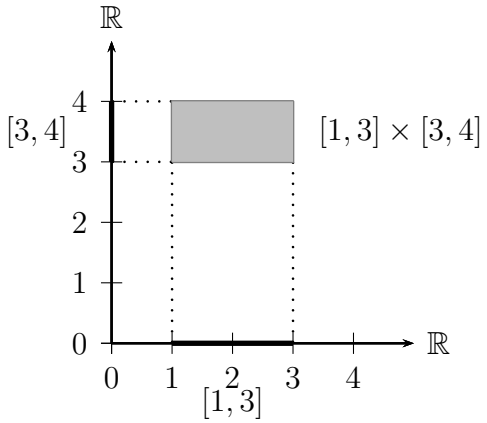
(7) $\{x \in A : P(x)\}$ where A is a set and $P(x)$ some property
Meaning: The set of elements x of A that satisfy property $P(x)$
Examples: $\{x \in \mathbb{R} : x \geq 0\}$ is the set of non-negative real numbers; $\{x \in \mathbb{R} : x^2 = 1\}$ is the set $\{-1, 1\}$

(8) $A \times B$, where A and B are sets

Meaning: $A \times B$ is called the Cartesian product of A and B, it is the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$.

Examples:

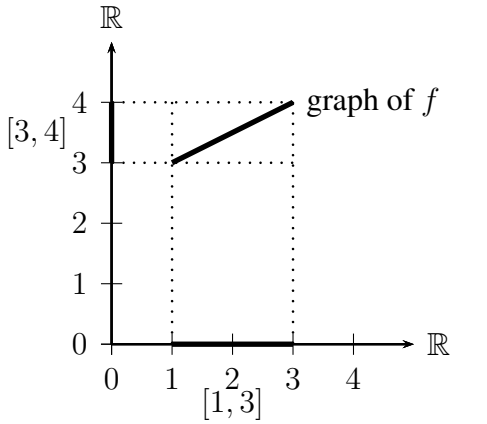
- $\mathbb{N} \times \mathbb{R}$ is the set of ordered pairs (n, r) such that n is a natural number and r is a real number;
- $\mathbb{R} \times \mathbb{R}$ is the set of ordered pairs (x, y) such that both x and y are real numbers;
- $[1, 3] \times [3, 4] = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \in [1, 3] \text{ and } y \in [3, 4]\}$ is the set depicted below



- if function f is defined by $f : [1, 3] \to [3, 4], x \mapsto \frac{1}{2}x + 2.5$ (see the beginning of our Slides 1 for the definition of a function), then the graph of f is the set

$$\{(x, y) \in [1, 3] \times [3, 4] : y = f(x) = \frac{1}{2}x + 2.5\},$$

which is a subset of the Cartesian product $[1, 3] \times [3, 4]$.



(9) A^2, where A is a set
Meaning: $A^2 = A \times A$
Example: $\mathbb{R}^2$ is the Cartesian product of $\mathbb{R}$ and $\mathbb{R}$, that is, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$

(10) A^m, where A is a set and $m \in \{1, 2, 3, \ldots...\}$
Meaning: $A^m = \underbrace{A \times A \times \ldots \times A}_{m \text{ times}}$
Example: $\mathbb{R}^4$ equals $\mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ and is the set of all quadruples (x_1, x_2, x_3, x_4) of real numbers

(11) $A \backslash B$, where A and B are sets
Meaning: set difference, $A \backslash B = \{x \in A : x \notin B\}$
Example: $\mathbb{R}^2 \backslash \{(0,0)\} = \{(x,y) \in \mathbb{R}^2 : (x,y) \neq (0,0)\}$

(12) $\mathbb{R}_+$
Meaning: $\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$

(13) $\mathbb{R}_{++}$
Meaning: $\mathbb{R}_{++} = \{x \in \mathbb{R} : x > 0\}$

(14) $\equiv$
Meaning: "is equal to," with the connotation that the the expression on the left-hand side of "$\equiv$" is defined by the right-hand side of "$\equiv$." Some authors write $\stackrel{def}{=}$ or also $:=$ instead of $\equiv$.
Example: Say $N_1 = 1$ and $N_2 = 3$. Then $N \equiv \max\{N_1, N_2\}$ means that we define N as $\max\{N_1, N_2\}(= 3)$, or that we let N equal $\max\{N_1, N_2\}(= 3)$.

(15) x, $\hat{x}$, $\tilde{x}$, x'
Meaning: These are just "names" for different variables. Instead of x, $\hat{x}$, $\tilde{x}$, x' we could equally well use x, w, y, z.
Example: Let $\varepsilon = 2$ and $\varepsilon' = \frac{\varepsilon}{2}$. Furthermore, let $x = 1$ and $\hat{x} = 2$. Finally, let $\tilde{x} = \varepsilon x$ and $x' = \varepsilon' \hat{x}$. Then $\tilde{x} = x' = 2$.
Note: "$'$" is also used with a different meaning, to indicate the derivative of a function of one variable. It is usually clear from the context which meaning "$'$" has. E.g., in "for $f : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$ and $x = 2$, $x' = 3$, the statements $f'(x) = 4$ and $f'(x') = 6$ are true," x' is a variable (which equals 3), while f' denotes the derivative of f.

(16) $|\cdot|$
Meaning: absolute value; the absolute value $|x|$ of a number $x \in \mathbb{R}$ equals $x \in \mathbb{R}$ itself if

$x \geq 0$, and $-x$ if $x < 0$. Note that $|x| \geq 0$ for all $x \in \mathbb{R}$.
Examples: $|3| = 3$; $|-10| = 10$